

Thursday Warm-up: Magnetic Fields and Work, Continuity Equation

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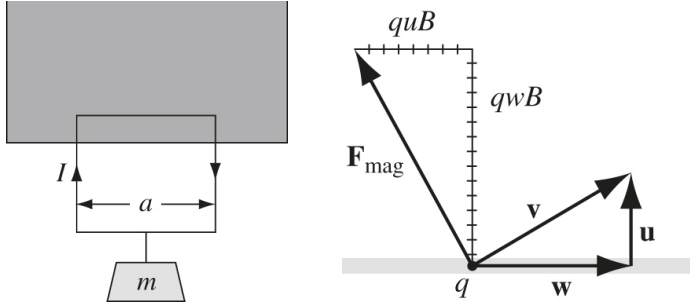


Figure 1: (Left) A loop of wire passes through a region with $\mathbf{B}$ into the page, supporting a mass m . (Right) The velocity components of the charge once the loop begins to move.

1 Memory Bank

- **Lorentz force** on a constant current I of length $\mathbf{L}$:

$$\mathbf{F} = I\mathbf{L} \times \mathbf{B} \quad (1)$$

- **Mechanical work:**

$$W = \int \mathbf{F} \cdot d\mathbf{l} \quad (2)$$

- **Continuity equation:**

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t} \quad (3)$$

2 Lorentz Force

1. Using the definitions in the Memory Bank, prove that magnetic fields do no work.

2. A rectangular loop of wire, supporting a mass m , hangs vertically with one end in a uniform magnetic field $\mathbf{B}$, which points into the page in the shaded region of Fig. 1 (left). What current I would exactly balance the gravitational force downward?

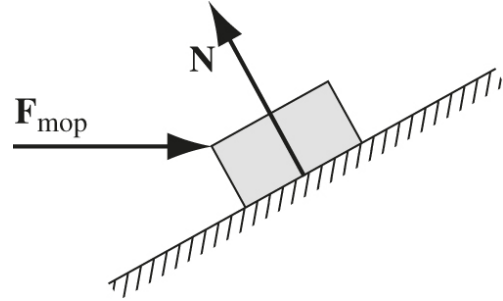


Figure 2: Applied force, normal force, on an incline.

3. If the current is *increased* from the value that balances the weight, the mass *rises*.¹ If the mass rises a distance h , work is being done against gravity. Write down the product of the Lorentz force and h .
4. Suppose the loop rises with constant velocity $\mathbf{u} = u\hat{\mathbf{y}}$, and the current is $I = \lambda w\hat{\mathbf{x}}$. (a) Show that the actual velocity of charge in the wire is $\mathbf{v} = \mathbf{u} + \mathbf{w}$. (b) Derive the components of the net force on the charge q .
5. Let the total charge on the line segment a be $q = \lambda a$. (a) Show that the vertical and horizontal components of the Lorentz force in Fig. 1 (right) are λawB , and λauB , respectively. (b) What work must be done against the horizontal force in a time dt to keep the current constant? That is, evaluate:

$$W = \int F_{\text{horiz}} dx = \int F_{\text{horiz}} w dt \quad (4)$$

Show this work is equal to the work required to lift the mass.

¹Students in PHYS180 did this experiment! It's cool to watch.